

AI for Fight The Landlord

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Abstract

Fight the Landlord is a challenging imperfect-information card game with a large combinatorial action space, hidden hands, and strong role-dependent cooperation and competition. In this project, we study several reinforcement learning and search-based approaches for building agents for this game. We first investigate value iteration and tabular value representations, but find that direct state enumeration is ineffective because the observable state must include not only the player's hand, but also public history, remaining card counts, turn context, and uncertainty over hidden cards. We then explore approximate Q-learning with history-aware features, neural policy learning through self-play, and adversarial search based on sampled determinizations of hidden hands. Finally, we combine learned Q-value estimators with adversarial search by using a trained Q model as the leaf evaluator. Our experiments and implementation process show that pure offline planning is difficult in this domain, while feature-based learning and bounded search provide more practical approximations for decision making under hidden information.

State Space and Imperfect Information Motivation

Introduction

Fight the Landlord, also known as Dou Dizhu, is a three-player card game with one landlord playing against two farmers. Although the rules are easy to describe, the decision problem is difficult for an artificial agent. A player must choose among many legal card combinations, reason about the hidden hands of the other two players, and adapt its strategy according to whether it is the landlord, the landlord's upper farmer, or the landlord's lower farmer. The game therefore combines a large action space, imperfect information, and team-based competition.

From a reinforcement learning perspective, the main difficulty is that the true game state is not fully observable. A player only knows its own hand, the public action history, the last played move, the number of remaining cards for each player, and its role. The exact cards held by the other players are hidden. As a result, a naive Markov decision process over complete hands is not directly usable at decision time, while a state representation based only on the current player's hand loses too much information about the actual game context.